

3.4.3 Relational, Logical and Membership Operators

ریشنل آپریٹرز (Relational Operators) دو ویلیوز کا موازنہ کر کے True یا False دیتے ہیں۔

CS XI

Chapter 3: Programming Fundamentals

Lecture 11 of 24

★ Golden

3.4.3 Relational, Logical and Membership Operators ★ Golden

LEARN IT

3. Relational (comparison) operators compare two values and return a Boolean (True or False). They are used in conditions.

Op	Description	Example (a = 10, b = 5)	Result
==	Equal to	a == b	False
!=	Not equal to	a != b	True
>	Greater than	a > b	True
<	Less than	a < b	False
>=	Greater than or equal to	a >= 10	True
<=	Less than or equal to	a <= b	False

```
a = 10
b = 5
print(a == b)
print(a != b)
print(a > b)
print(a < b)
print(a >= 10)
print(a <= b)
```

```
OUTPUT
False
True
True
False
True
False
```

4. Logical operators combine several conditions:

A	B	A and B	A or B	not A
False	False	False	False	True
False	True	False	True	True
True	False	False	True	False
True	True	True	True	False

- and — True only if both conditions are true.
- or — True if at least one condition is true.
- not — reverses the result (True becomes False, False becomes True).

```
a = 10
b = 5
print(a > 5 and b > 5)
print(a > 5 or b > 5)
print(not a > 5)
```

```
OUTPUT
False
True
False
```

5. Membership operators check whether a value exists inside a sequence (such as a string or list): in and not in.

```
name = "Python"
print("P" in name)
print("z" not in name)
print("p" in name)
marks = [85, 98, 75]
print(98 in marks)
```

```
OUTPUT
True
True
False
True
```

"P" in name is False because strings are case-sensitive: the word has a capital P.

WORKED EXAMPLE

Real-life conditions. A person can apply for a CNIC if they are 18 or older and have the documents. Marks are invalid if they are below 0 or above 100.

```
age = 20
has_cnic = True
print(age >= 18 and has_cnic)
marks = 45
print(marks < 0 or marks > 100)
```

```
OUTPUT
True
False
```

Step by step: 20 >= 18 is True, has_cnic is True, True and True = True. For marks: 45 < 0 is False, 45 > 100 is False, False or False = False (the marks are valid).

REMEMBER IT

AND is strict, OR is generous. AND is like a teacher who needs both homework and attendance; OR is like a shop that accepts cash or card. Order of evaluation: not first, then and , then or .

COMMON MISTAKE

Do not confuse = (assignment, "store") with == (comparison, "is it equal?"). Writing if x = 5: is a syntax error; the correct form is if x == 5: .

اردو میں سمجھیں

ریشنل آپریٹرز (Relational Operators) دو ویلیوز کا موازنہ کر کے True یا False دیتے ہیں۔ لاجیکل آپریٹرز میں and , True دیتا ہے جب دونوں شرطیں درست ہوں، or , جب کم از کم ایک درست ہو، اور not نتیجہ الٹ دیتا ہے۔ in اور not in دیکھتے ہیں کہ کوئی چیز اسٹرینگ یا لسٹ میں موجود ہے یا نہیں۔

EXAM TIP

Typical: "Evaluate a > b and b > 3 for a = 10, b = 5" or "Write the truth table of and / or ". Write each sub-condition's value first, then combine. Remember: == compares, = assigns.

CHECK YOURSELF

1. Predict the output:

```
print(True and False or True)
print(not (5 > 3) or 2 == 2.0)
```

2. Find the error:

```
x = 5
if x = 5:
    print("five")
```

Answers: 1. True and True . (True and False = False; False or True = True. not True = False; 2 == 2.0 is True; False or True = True.) 2. = is assignment; a comparison needs == . Python reports SyntaxError: invalid syntax . Maybe you meant '==' or ':=' instead of '='? Fix: if x == 5:

Next lecture: 3.4.3 Bitwise Operators